LECTURE 21

Random Forests

Using a bunch of overfitters to balance the Bias-Variance trade-off

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Content credit: Acknowledgments

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Random Forests

Multiclass Classification

Decision Trees (conceptually)

- Creating Decision Trees in `sklearn`
- Evaluating Tree Accuracy
- Overfit Decision Tree Example

The Decision Tree Generation Algorithm

- Evaluating Split Quality
- Entropy
- Generating Trees Using Entropy

Avoiding Overfitting

- Restricting Decision Tree Complexity
- **Random Forests**

Summary and Context

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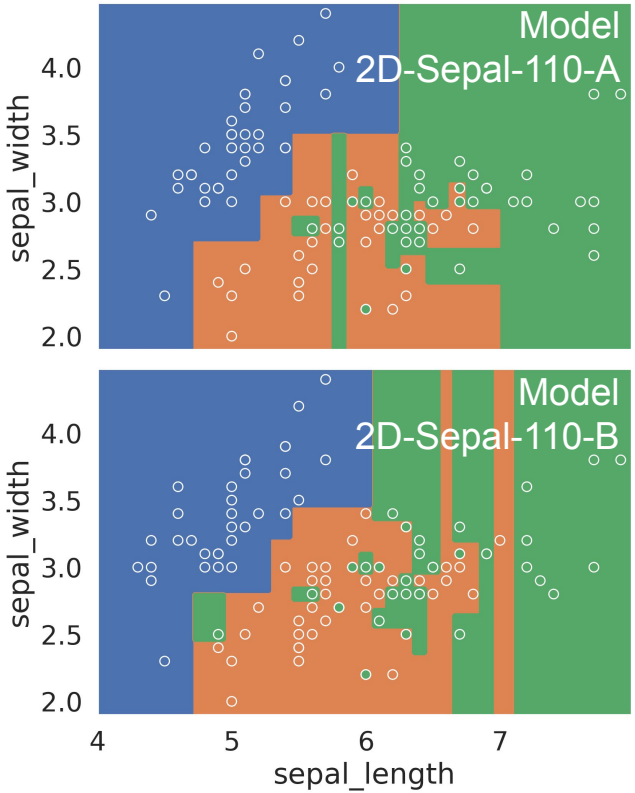
Random Forests: Harnessing Variance

As we've seen, fully-grown decision trees will almost always overfit data

- Low model bias, high model variance.
- In other words, small changes in dataset will result in very different decision tree.
- **Example**: Two models on the right trained on different subsets of the same data.

Random Forest Idea: Build many decision trees and have them vote.



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Random Forests: Harnessing Variance

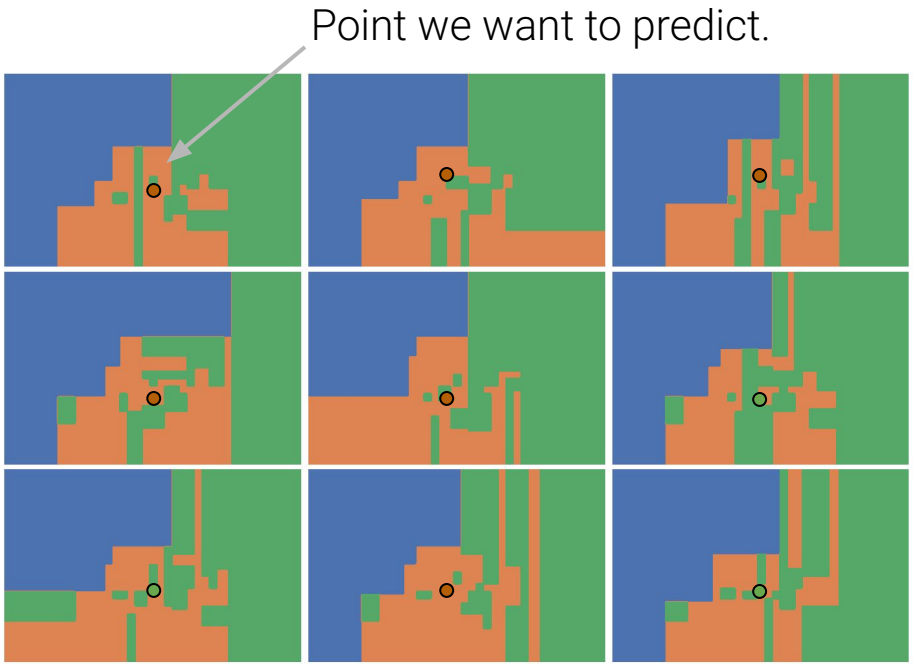
What do we mean by vote?

- For a given x/y, use whichever prediction is most popular.

Consider example at right with 9 models.

- 6 votes orange, 3 votes green.
- Random forest prediction is orange.

This kind of methods is called **ensemble learning**.



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Building Many Trees

Big fundamental problem: We only have one training set.

How can we build many trees using one training set?

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Bagging

Bagging: Short for **B**ootstrap **AGG**regat**ING**.

- Generate bootstrap resamples of training data.
- Fit one model for each resample.
- Final model = average predictions of each small model.
- Invented by Leo Breiman in 1994 (Berkeley Statistics!).

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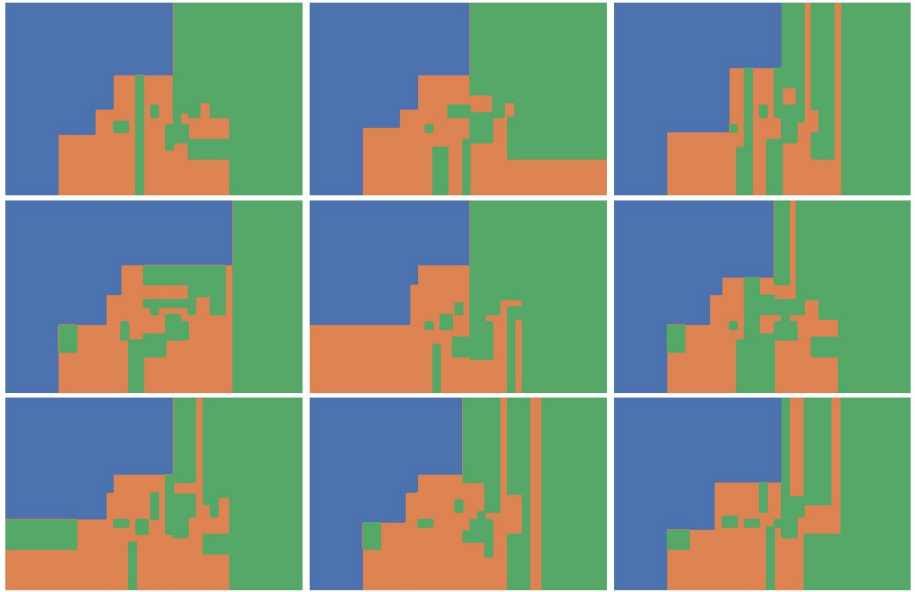
Measuring a Tree: Out-of-Bag (OOB) Prediction Error

- When we bootstrap a sample, we exclude some of the values.

- To test how well a tree generalizes, we want to test it's predictions on unseen values.

OOB: Predict the values that weren't in the bootstrap!



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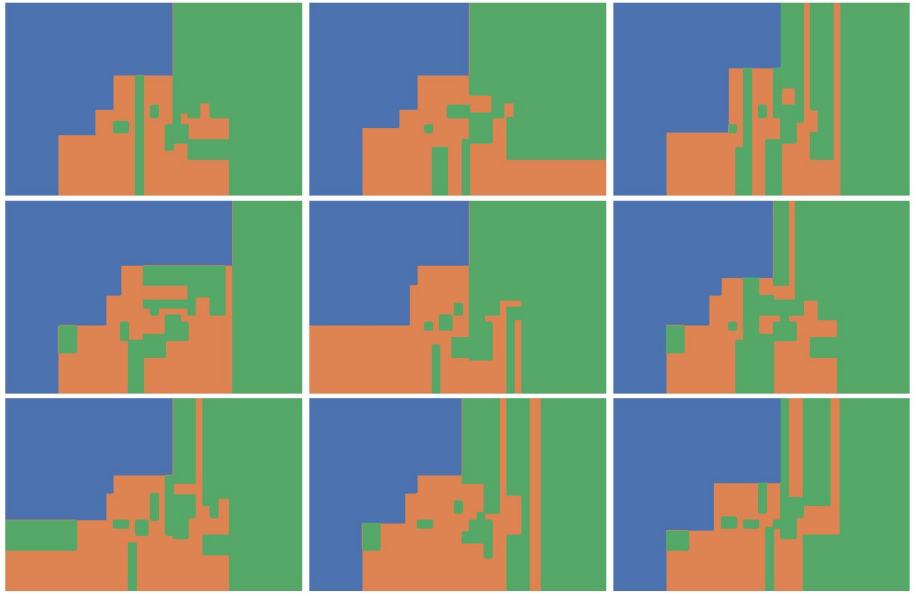
Random Forests

- Bagging often isn't enough to reduce model variance!
  - Decision trees often look very similar to each other and make similar predictions.
  - Ensemble will still have low bias and high model variance.
  - E.g., one strong feature always used for first split.



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Random Forests

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- **Idea:** Only use a random subset of ***m*** features at each split.
  - Usually $m = \sqrt{p}$ for decision trees used for classification.
  - Here p is the number of features.
- Algorithm creates individual trees, each overfit in a different way.
  - The hope is that the overall forest has low variance.

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Random Forests Algorithm

- *Bootstrap training* data T times. For each resample, fit a decision tree by doing the following:
  - Start with data in one node. Until all nodes **pure** or **unsplittable**:
    - Pick an impure node.
    - *Pick a random subset of m features.* Pick the ***best feature*** x and ***best split value*** β such that the **weighted entropy** of the resulting split is minimized
    - Split data into two nodes, one where $x \leq \beta$, and one where $x > \beta$
- To predict, ask the T decision trees for their predictions and take majority vote.

This approach has two hyperparameters T and m.

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Variable Importance

In a linear (logistic) model with standardized features, the coefficients tell us how important a feature is.

Random forests don't have coefficients, but gain the following idea:

- At each split, predictors are sampled at random.
  - The strongest predictor is chosen.
- Across many trees, we can record whether a feature was used.

Variable Importance: How often was a feature chosen as the strongest?

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Avoiding Overfitting with Heuristics

We've seen many approaches to avoid overfitting decision trees.

- Preventing growth.
- Pruning.
- Random forests.

These ideas are generally "heuristic"

- Not provably best or mathematically optimal.
- Instead, they are just ideas that somebody thought sounded good, implemented, then found to work in practice "acceptably" well.

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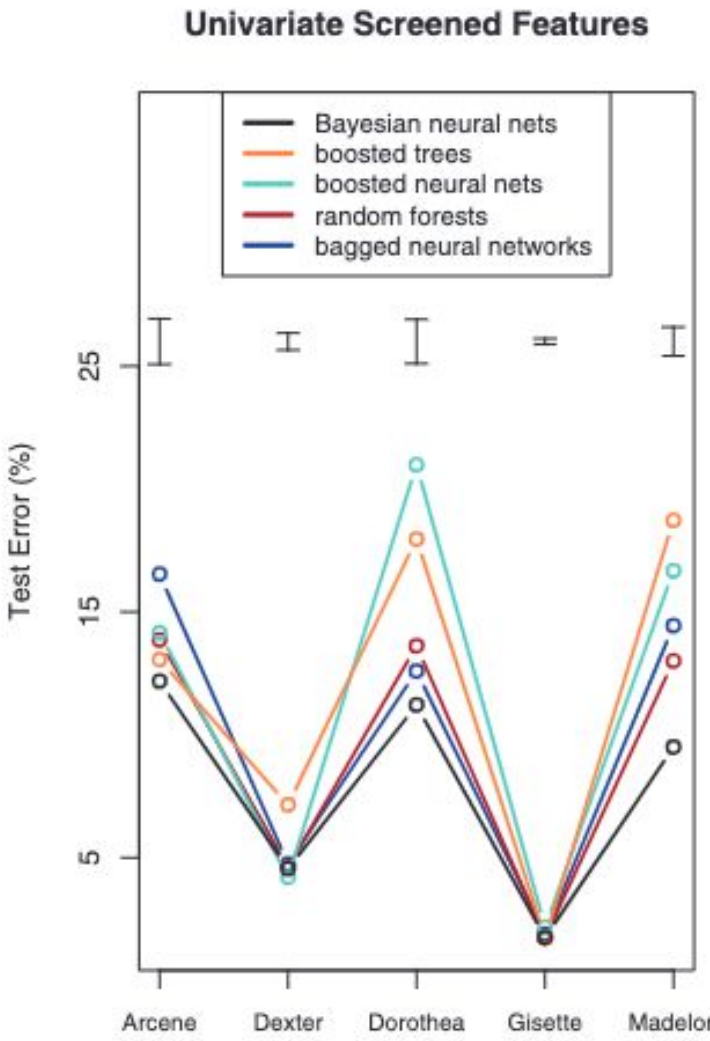
Fit "Acceptably" Well

The plot shows the results of NeurIPS 2003 challenge.

- x-axis shows 5 different data sets
- y-axis shows the test error for 5 different models
  - Each model represents *months* of work
- "Univariate Screened Features"
  - Lots of data pre-processing

The clear winner was Bayesian neural nets (black line).

- The random forests model, though...



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Hyperparameters

Only two! The number of trees and the number of predictors to try at each split!

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...

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...

Except we still have all of the Decision Tree parameters (including ones we haven't talked about):

- Number of trees (bootstrap resamples)
- Number of predictors, as well as
- Max depth of each tree
- Minimum samples to split
- Minimum impurity to split

In practice, we usually use CV for # trees and # predictors, and use the others at a default value.

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Avoiding Overfitting

- Restricting Decision Tree Complexity
- Random Forests

Summary and Context

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Decision Tree Generation

Decision tree generation algorithm:

- All of the data starts in the root node
- Repeat until every node is either **pure** or **unsplittable**:
  - Pick the ***best feature*** x and ***best split value*** β such that the **weighted entropy** of the resulting split is minimized, e.g. x = `petal_width`, β = 0.8 has loss 0.66
  - Split data into two nodes, one where $x \leq \beta$, and one where $x > \beta$

Notes: A node that has only one type is called a "**pure**" node. A node that has duplicate data that cannot be split is called "**unsplittable**"

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Why Decision Trees?

Decision Trees present an alternative philosophy to logistic regression.

- Instead of a weighted sum of features, we use a series of yes/no questions.

Some advantages of a decision tree:

- Able to capture nonlinear relationships (at risk of overfitting).
- Output is more interpretable: You know exactly how the tree made a decision.
  - Example: Imagine trying to make an argument in a court case.
  - This is not true of random forests.

Some disadvantages:

- Care needs to be taken to avoid overfitting.
- Not as mathematically nice (e.g. Ridge Regression has a simple closed form solution).

If all you care about is accuracy (or precision) (or recall), use **cross validation** to pick between the two modeling philosophies.

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Random Forests Algorithm

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- To predict, ask the T decision trees for their predictions and take majority vote.

This approach has two hyperparameters T and m (but there are others that we often ignore).

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Why Random Forests?

- Versatile: does both regression and classification.
- Invariant to feature scaling and translation.
- Automatic feature selection.
- Nonlinear decision boundaries without complicated feature engineering.
- Doesn't overfit as often as other nonlinear models (e.g. polynomial features).
- Example of **ensemble method**: combine the knowledge of many simple models to make a sophisticated model.
- Example of using **bootstrap** to reduce model variance.

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Participation Questions

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Q1

Which of the following does *not* change if the units of a feature are changed (e.g. using mm instead of inches):

A. The coefficients in a linear model

B. The coefficients in a logistic regression model

C. The optimal decision split point

D. The loss at the optimal split point

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Q2

Random forests are made from a bunch of decision trees, which tend to overfit. Therefore, we can expect random forests to overfit.

A. TRUE

B. FALSE

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Q3

Which of the following is not chosen via cross-validation?

A. The number of trees used

B. The number of parameters used

C. The optimal predictor at each split

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Q4

Which of the following concepts is *not* used in Random Forests?

A. Regularization

B. Cross-validation

C. Inference

D. Bootstrapping

E. The bias-variance trade-off

F. Minimizing a loss function

G. Prediction

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